Nguyen Song Thien Long

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Research Interests

My research asks how retrieval-augmented language-model systems can reason reliably over high-stakes, specialized knowledge in languages with limited resources. I pursue this in **Natural Language Processing (NLP)**, focusing on Information Retrieval (IR), Question Answering (QA), and Large Language Models (LLMs) for Vietnamese and ethnic minority languages. My work approaches it from three angles: benchmarks that evaluate reasoning in specialized domains, agentic systems that remain trustworthy under uncertainty, and methods that adapt retrieval and translation models to new languages. I apply this work in domains such as law, healthcare, and education, aiming for methods that make language technology work for underserved communities.

Employment

Apr 2026 – Present
Principal Investigator. Saigon AI Hub, VNG Group.
Leading a granted research project on structured reasoning and training strategies for LLMs, funded by VNG Group and Vietnam National University Ho Chi Minh City (VNU-HCM).

Oct 2025 – Present
AI Researcher. Center for AI Research (CAIR), VinUniversity.
Investigating the training of lexical–dense and hybrid retrieval models and their integration into LLM-based QA systems, mentored by Prof. Luu Anh Tuan (NTU, Singapore) and Dr. Nguyen Tran Cong Duy (CAIR).

Dec 2023 – Present
Research Assistant. URA Research Group, Faculty of Computer Science and Engineering, HCMUT.
Conducting research on IR, QA, and LLM-based systems under the supervision of Prof. Quan Thanh Tho, and serving as *Team Leader*, defining research directions and mentoring junior researchers.

Education

2023 – 2027
B.Eng. in Computer Science (High-Quality Program), Ho Chi Minh City University of Technology (HCMUT), VNU-HCM, Vietnam. CGPA: 3.6/4.0.

Selected Publications (full list at https://longstnguyen.com/publications/)

2027
- **Long S. T. Nguyen**. *English-Only Adaptation Splits Monolingual and Cross-Lingual Retrieval: Task-Axis Dissociation and Retention Trade-offs.* ACM International Conference on Web Search and Data Mining (**WSDM**). (CORE A)
- **Long S. T. Nguyen**, Cong-Duy Nguyen, Anh Tuan Luu, Vu Duong, and Tho T. Quan. *Sentence Matching Is Not Enough: Relational Geometry Transfer for Post-Hoc Language Expansion of Retrievers.* ACM International Conference on Web Search and Data Mining (**WSDM**). (CORE A)
- Thi Ty Nguyen, **Long S. T. Nguyen**, Tan Sang Nguyen, Nhu Tran, Khang Nguyen, Dong Doan, Vy Le, Ngoc Tran, Phat T. Tran-Truong, and Tho T. Quan. *Fluent but Foreign: Benchmarking Cultural Understanding of Multilingual Large Language Models in Vietnamese.* ACM International Conference on Web Search and Data Mining (**WSDM**). (CORE A)

2026
- **Long S. T. Nguyen**, Dat T. Huynh, Quynh T. N. Vo, Dang Le, Cong-Duy Nguyen, Anh Tuan Luu, Anh Nguyen-Duc, and Tho T. Quan. *Fast and Focused: Interactive Repository-Level Fill-in-the-Middle Code Completion with Hybrid Structural Retrieval.* Conference on Empirical Methods in Natural Language Processing (**EMNLP**). (CORE A*)
- **Long S. T. Nguyen** and Tho T. Quan. *Which Works Best for Vietnamese? A Practical Study of Information Retrieval Methods across Domains.* Findings of the Association for Computational Linguistics: **EACL**. (CORE A)
- Thi Ty Nguyen, **Long S. T. Nguyen**, Phat T. Tran-Truong, Tan Sang Nguyen, and Tho T. Quan. *Bahnar–Vietnamese Cross-Lingual Sentence Retrieval: A Low-Resource Case Study with Lexical, Neural, and Hybrid Methods.* Asia-Pacific Chapter of the Association for Computational Linguistics & International Joint Conference on Natural Language Processing (**AACL-IJCNLP**).
- Hung Luu, **Long S. T. Nguyen**, Trung Pham, Hieu Pham, and Tho Quan. *HiGraAgent: Dual-Agent Adaptive Reasoning over Hierarchical Knowledge Graph for Open-Domain Multi-hop Question Answering.* Findings of the Association for Computational Linguistics: **EACL**. (CORE A)
- Quoc Phu Dang*, Phat T. Tran-Truong*, Duc-Ly Vu, **Long S. T. Nguyen**, Quynh T. N. Vo, and Tho T. Quan. *Enhancing Large Language Model Performance for Automatic Zero-Shot Multiple-Choice Question Answering via Single-Token Logit Prompting.* Computers and Education: Artificial Intelligence. (SCIE, Q1, #1 IN AI)

2025
- **Long S. T. Nguyen**, Quynh T. N. Vo, Hung C. Luu, and Tho T. Quan. *When in Doubt, Ask First: A Unified Retrieval Agent-Based System for Ambiguous and Unanswerable Question Answering.* Findings of the Association for Computational Linguistics: **IJCNLP-AACL**.
- **Long S. T. Nguyen**, Tran T. B. Le, Huong P. N. Nguyen, Quynh T. N. Vo, Phong H. N. Nguyen, and Tho T. Quan. *Serving the Underserved: Leveraging BARTBahnar Language Model for Bahnaric-Vietnamese Translation.* Workshop on Language Models for Underserved Communities (**LM4UC**), at **NAACL**. (WORKSHOP, CORE A)

Academic Service

Prog. Cmte. AAMAS 2027 (CORE A); AAAI/EAAI 2027 (CORE A*); ACM WSDM 2027 (CORE A).

Conf. Rev. WWW 2027 (CORE A*); ACM MM 2026 (CORE A*).

Academic Service (continued)

Journal Rev. — IEEE Transactions on Neural Networks and Learning Systems (Q1); Expert Systems with Applications (Q1); Engineering Applications of Artificial Intelligence (Q1); IEEE Access (Q1); Behaviour & Information Technology (Q1); Journal of Natural Language Processing (Q1).

Olympiad — Technical and training support for VOAI 2025 and the preparation of Vietnam's IOAI 2025 team.

References

Supervisor — **Prof. Quan Thanh Tho.** Dean, Faculty of Computer Science and Engineering, HCMUT. qttho@hcmut.edu.vn

Mentor — **Prof. Luu Anh Tuan.** Nanyang Technological University (NTU), Singapore. anhtuan.luu@ntu.edu.sg

Prof. Duong Nguyen Vu. Vice Provost of Graduate Education & Chief Scientific Officer, CAIR, VinUniversity. vu.dn@vinuni.edu.vn

Dr. Nguyen Tran Cong Duy. Postdoctoral Researcher, CAIR, VinUniversity. duy.ntc@vinuni.edu.vn